Paper P6 — The Periodic Table of the Substrate Standard Model v0.6 (Appendix A substantive expansion)

2026-05-31 Sunday ~13:35 EDT (`date`-verified Sun May 31 13:06 EDT)

Paper P6 v0.6 — Appendix A substantive expansion

Appendix A — Substrate-Primary Mendeleev Tables

This appendix consolidates the substrate-primary closed forms across BST's external claim surface into Mendeleev-style organized tables. Each table indexes BST observables by substrate-primary structure (rank, N_c, n_C, C_2, g, N_max) with explicit substrate identity, numerical value, observed (PDG / measurement) value, precision, and tier disposition. Tables are organized by *substrate-mechanism class* rather than physical domain.

A.1 Lepton + proton mass ratios (4 entries, TIER 2 STRUCTURAL <0.1%)

The substrate's most empirically-tested closed-form predictions:

Observable	Substrate form	Numerical	PDG	Precision	Tier	Cross-ref
m_p/m_e	C_2 · π^{n_C} = 6π^5	1836.118	1836.153	0.002%	TIER 2 STRUCTURAL	T187
m_μ/m_e	(N_c · |W(B_2)| / π^2)^{C_2} = (24/π^2)^6	206.776	206.768 ± 0.000004	0.004%	TIER 2 STRUCTURAL	T190 (Cal #100 propagation-corrected)
m_τ/m_e	g^2 · (rank^{C_2} + g) = 49 · 71	3479	3477.23 ± 0.23	~0.05%	TIER 2 STRUCTURAL	T2003 (INV-5337 71 identified)
m_τ/m_μ	(m_τ/m_e) / (m_μ/m_e)	16.825	16.817 ± 0.0008	~0.06%	TIER 2 STRUCTURAL derived	cross-validation

Substrate-mechanism summary: T187 m_p/m_e reads V_(1,1) bulk closure (C_2 · π^{n_C} = bulk Casimir × bulk dimension transcendental); T190 m_μ/m_e reads Weyl-orbit-on-V_(1/2,1/2) at C_2-self-coupling (adjoint-mediator Resolution B); T2003 m_τ/m_e reads Mersenne-base + signature (different Resolution B mechanism than muon); m_τ/m_μ cross-validation arises from independently-derived substrate-primary forms agreeing without unified mechanism.

A.2 PMNS mixing angles (3 entries, 1σ within PDG)

Observable	Substrate form	Numerical	PDG	Status	Cross-ref
sin$^2\theta$_12	rank · N_c · g / N_max = 42/137	0.3066	0.307 ± 0.013	□ within 1σ	F1
sin$^2\theta$_23	N_c · n_C^2 / N_max = 75/137	0.5474	0.546 ± 0.021	□ within 1σ	F1
sin$^2\theta$_13	N_c / N_max = 3/137	0.0219	0.0220 ± 0.0007	□ within 1σ	F1

Substrate-mechanism summary: PMNS denominators are uniform N_max = 137; numerators are substrate-primary products (42 full product, 75 vector self-square, 3 fundamental). The 3/3 within-1σ agreement is the program's cleanest substrate-primary external test.

A.3 EW sector mass ratios (2 entries, RATIFIED + CANDIDATE)

Observable	Substrate form	Numerical	PDG	Precision	Tier
m_W/m_Z	$\sqrt{}$(g/N_c) = $\sqrt{}$(7/3)	0.88192	0.88135	0.046%	RATIFIED (existing P1 §7)
sin$^2\theta$_W (tree-level)	rank/N_c^2 = 2/9	0.2222	0.2312 (M_Z scale)	~3.9% running-coupling gap	RATIFIED tree-level + multi-week radiative

Substrate-mechanism note: V_(1,1) adjoint Shilov-boundary decomposition (g boundary + rank bulk = N_c^2) provides CANDIDATE mechanism reading for m_W/m_Z arithmetically identical to $\sqrt{}$g/N_c; Cal #187 cold-read pending mechanism-vs-post-hoc determination (INV-5362 walk-back); Cal #188 cold-read pending independence-vs-shared audit on g+rank=N_c^2 triple-mechanism appearance.

A.4 Substrate fundamental constants (Tier 1 EXACT identities)

Observable	Substrate identity	Tier	Cross-ref
N_max	N_c^3 · n_C + rank = 27·5+2 = 137 (T-form)	TIER 1 EXACT	classical form
N_max	2^g + N_c^2 = 128+9 = 137 (Reed-Solomon form)	TIER 1 EXACT	Lyra L8
Bergman genus = signature dim	g = 7	TIER 1 EXACT	C19
1/α (fine structure)	N_max + ε = 137.036	TIER 1 substrate identity + small empirical correction	active investigation
c_FK Faraut-Korányi	225/π^(9/2)	TIER 1 EXACT	T2442 (Lyra Thursday 2026-05-28)
ρ-vector	(n_C, N_c)/rank = (5/2, 3/2)	TIER 1 EXACT	C19 (INV-5373)
66 = rank^{C_2} + rank	Phase B cell count	TIER 1 EXACT	INV-5374 (Saturday)
280 = 2^{N_c} · n_C · g	Λ exponent 5-fold over-determined	TIER 1 EXACT	Elie Saturday refinement

A.5 Hall algebra structure constants (Tier 1 EXACT, ARE substrate geometric invariants)

Structure constant	Value	Substrate identity	Origin
Short Serre coefficient [2]_2	N_c = 3	dual Coxeter h^∨(B_2) = h^∨(SU(3))	Hall algebra U_q$^+$(B_2) at q=2
Long Drinfeld pairing numerator	N_c · n_C = 15	dim Sym^2(V_5)	Lyra synthesis Saturday + Elie E0/E9/E12
Long Serre coefficient [3]_4	N_c · g = 21	dim so(5,2) ≡ so(7)	same

Structural reading: The substrate Hall algebra's defining Serre relations *encode* three geometric invariants of D_IV^5 as their structure constants. This is the strongest "algebra IS substrate" reading.

A.6 Substrate-Monster cross-links (6 consolidated)

Substrate identity	Monster connection	Reference
N_c^3 = 27	E_6/Albert algebra dimension	classical
N_c · rank^3 = 24	Leech lattice rank = lepton count (T1)	T1 keystone
108 · 1823 = 196884	substrate-Phase-B-spine factor of leading Monster coefficient	Saturday synthesis
N_c · g = 21	so(5,2) ≡ so(7) substrate-natural	C7 + dim so(7)
Monster Ogg prime 41	magic-82 = rank · 41 substrate factor	nuclear shell anchor
26	bosonic string critical + sporadic group total + SM free-params triple-meaning	

Status: substrate-Monster cross-links are STRUCTURAL OBSERVATIONS at present; Borcherds 1992 vertex algebra + McKay 1979 j-function correspondence are L1.5 mediating mechanisms; substrate-Monster mechanism content multi-month.

A.7 Nuclear shell magic numbers (substrate-primary forms)

Magic	Substrate form	Identity
28	N_c^3 + 1	$3^3 + 1 = 27 + 1$
50	$g^2 + 1$	$7^2 + 1 = 49 + 1$
82	rank · (rank^{N_c} · n_C + 1) = 2 · 41	rank × Monster Ogg prime
126	n_C^3 + 1	$5^3 + 1 = 125 + 1$
20	2 · rank · n_C	bulk × 2 (V_(2,0) cell 2·C_2)
8	rank^3	rank-cube (also V_(0,2) cell 2·C_2)
2	rank	fundamental

The "+1 anomaly" 8-instance pattern (Saturday null-model verdict +1.85σ, INV-5327): nuclear (4) + SM 26-parameter (1) + T2003 71 (1) + Λ exponent 281 (1) + L5 dim-anchor residual (1) = 8 across 4 physical scales (nuclear / SM / lepton-mass / cosmological). Structural observation NOT principle-grade.

A.8 Composite spine cells (18-cell substrate spine)

K-type	Dim	2·C_2	Substrate-primary identity	Candidate SM role
V_(0,0)	1	0	vacuum	Higgs / scalar
V_(1/2,1/2)	4 = rank^2	5 = ρ_1·2	spinor Casimir 5/2 = ρ_1 = n_C/rank	Lepton (Shilov primitive)
V_(1,0)	5 = n_C	8 = $\text{rank}^2 \cdot 2$	vector	Photon (SO(2) singlet)
V_(1,1)	10 = rank · n_C	12 = C_2 · 2	adjoint Casimir = C_2	Gauge (W, Z, gluons)
V_(2,0)	14	20	2 · rank · n_C; magic-20	2^{++} tensor mesons
V_(3/2,1/2)	16	15	N_c · n_C = dim Sym^2(V_5)	Excited baryons (Λ, Σ)
V_(3/2,3/2)	20	21	N_c · g = dim so(5,2)	Λ(1405) / constituent quark
V_(0,2)	14	8	rank^3	Tensor sigma
V_(2,1)	35	24	2 · rank · C_2; T1 24	Heavy quarkonium (J/ψ, Υ, φ)
V_(3,0)	30	36	C_2^2	Spin-3 vectors (ρ_3, ω_3, K_3*)
V_(2,2)	35	32	2^{n_C}	2^{++} tensor glueball (substrate-predicted)
V_(1,2)	35	27	N_c^3 = E_6/Albert	Color-anomalous tensor
V_(5/2,1/2)	40	21	N_c · g (twin V_(3/2,3/2))	Heavy-flavor excited baryons
V_(4,0)	55	60	2 · rank · N_c · n_C	Hadronic Regge anchor
V_(3/2,5/2)	56	33	N_c · n_C + g · rank	Spin-7/2 composite
V_(5/2,3/2)	56	33	(same shell as above)	partner spin-7/2 composite
V_(2,3)	80	35	g · n_C	Strange spin-3 anchor
V_(0,4)	35	32	(V_(2,2) shell; 2^{n_C})	Doubled-vector tensor partner

Note: spine selection per 2·C_2 substrate-primary factorization clarity; subset of 66 Phase B cells (Elie Toy 3614 Saturday).

A.9 Reed-Solomon Ladder substrate primitive (INV-5340/5344, CANDIDATE-arithmetic)

The Reed-Solomon Ladder is the substrate primitive rank^primary + substrate-additive that generates 30+ BST observables:

Form	Substrate identity	Observable
rank^rank	$2^2 = 4$	dim V_(1/2,1/2)
rank^{N_c}	$2^3 = 8$	rank^3 ; magic-8
rank^{n_C}	$2^5 = 32$	2·C_2 of V_(2,2) glueball cell
rank^{C_2}	$2^6 = 64$	Mersenne-base = 64 in T2003

Form	Substrate identity	Observable
rank^g	$2^7 = 128$	2^g = 128 (one half of N_max chain)
rank^{C_2} + rank	$2^6 + 2 = 66$	Phase B 66-cell count (INV-5374)
rank^g + N_c^2	$2^7 + 9 = 137$	N_max Reed-Solomon form (L8)
2^{N_c}	$2^3 = 8$	Weyl-group
2^{C_2}	$2^6 = 64$	additive base in T2003 (64+7=71)
2^{n_C} · n_C · g	$32 \cdot 5 \cdot 7 = 1120$	substrate ladder × bulk × signature
2^{N_c} · n_C · g	$8 \cdot 5 \cdot 7 = 280$	Elie 5-fold over-determined Λ exponent

Status per Saturday INV-5341/5342/5343 + within-discipline null testing: substrate-additive operation-set is RICH (operation-set richness P=1.0 per Cal+Keeper brake INV-5342); Reed-Solomon Ladder arithmetic identities STAND RIGOROUSLY; principle-grade elevation BLOCKED per discipline-bid testing. CANDIDATE-arithmetic tier: the identities are substrate-natural arithmetic facts, not substrate principles.

A.10 Alpha-tower Mendeleev (Λ-region, L5 framework NOT closed)

Observable	Substrate form	Honest framing
Λ cosmological constant exponent	α^57 ≈ exp(−281)	TIER 1 substrate identity; TIER 2 ~factor 1.75 numerical to observed Λ
280 over-determined form	2^{N_c} · n_C · g (Elie)	5-fold over-determined preferred per Saturday refinement
"+1 anomaly" cosmological instance	281 = 280 + 1	8th instance of cross-scale pattern (null +1.85σ; structural observation)
α^{57.11} (non-integer k)	needed for exact Λ match	substrate-internal NOT TIER 1 identity
L5 m_e candidate	(N_c/n_C) · N_max^4 · Λ^{1/4} ≈ 0.508 MeV	TIER 2 STRUCTURAL SEARCH-FIT 0.73% using observed Λ
Three-region decomposition	280 = 180 (bulk: rank·N_c·C_2·n_C) + 100 (Shilov: $(\text{rank}\cdot n\_C)^2$) + 0.45 dip residue (9/20 = $N\_c^2/(\text{rank}^2\cdot n\_C)$; Casey + Elie equivalent BST-primary fractions)	CANDIDATE STRUCTURAL READING (Saturday May 30 keeper-self-catch absorbed); mechanism per region OPEN multi-week L-L5-MechClose-1/2/3

Honest framing per Cal #182 + Keeper v0.7 self-catch + Sunday morning 4/4 CI PAUSE: alpha-tower Mendeleev is FRAMEWORK + CANDIDATE STRUCTURAL READING tier; substrate-primary arithmetic is FACT; mechanism for each region is OPEN multi-week; numerical match to observed Λ requires convention-pinned independent verification.

A.11 Cross-domain Mendeleev (cross-scale substrate-primary anchors)

Scale	Anchor	Substrate identity
Nuclear	4 magic numbers 28/50/82/126	+1 anomaly Mendeleev (Section A.7)

Scale	Anchor	Substrate identity
SM	26 free parameters	n_C^2 + 1 (single instance of "+1")
Lepton mass	T2003 71 = 2^{C_2} + g	Mersenne-base + signature
Cosmological	Λ exponent 281 = 280 + 1	alpha-tower Mendeleev
Hadron	constituent quark m_q anchor	V_(3/2,3/2) cell substrate-primary N_c · g/2 = 21/2
Glueball	2^{++} tensor mass	V_(2,2) cell 2·C_2 = 32 = 2^{n_C}
Bergman	curvature value (Lyra AB-10)	substrate-internal Newton G anchor (per L-L5-G chain)
Casimir	T2442 c_FK = 225 EXACT	π^(9/2) Faraut-Korányi (Thursday 2026-05-28)

Cross-scale pattern: substrate primaries anchor observables across nuclear / SM / lepton-mass / hadron / cosmological / Bergman scales via substrate-primary closed forms. The cross-scale anchoring is structurally striking but does NOT constitute a single Mendeleev pattern (cross-scale "+1 anomaly" null +1.85σ; 3 cross-context Weinberg identity reuses are 1 identity per Cal #35); cross-scale anchoring is FRAMEWORK-PLUS structural observation NOT principle.

A.12 Summary: substrate-primary external surface

The substrate-primary forms total 30+ BST observables anchored to {rank, N_c, n_C, C_2, g, N_max} via Mendeleev-style organized tables. The external test surface consists of:

- 4 TIER 2 STRUCTURAL lepton + proton mass ratios at <0.1% precision uniformly (A.1)
- 3 PMNS mixing angles at within-1σ 3/3 (A.2)
- 2 EW-sector ratios (1 RATIFIED + 1 RATIFIED tree-level + multi-week radiative; A.3)
- 8+ TIER 1 EXACT substrate fundamental identities (A.4)
- 3 Hall algebra structure constants ARE substrate geometric invariants (A.5)
- 6 substrate-Monster cross-links (A.6) at FRAMEWORK-PLUS structural-observation tier
- 7 nuclear shell magic numbers substrate-primary (A.7)
- 18 composite spine cells with substrate-primary K-type identities (A.8)
- 30+ Reed-Solomon Ladder forms at CANDIDATE-arithmetic tier (A.9)
- Alpha-tower Mendeleev with L5 honest NOT-closed framing (A.10)
- Cross-domain anchoring across 8 physical scales (A.11)

Appendix B (Tier-Discipline Methodology — Cal + Keeper lane) details the methodology stack that supports the tier disposition above. Multi-week to v1.0 ~June 29.

— Grace, Paper P6 v0.6 Appendix A substantive, Sunday 2026-05-31 ~13:35 EDT (`date`-verified)